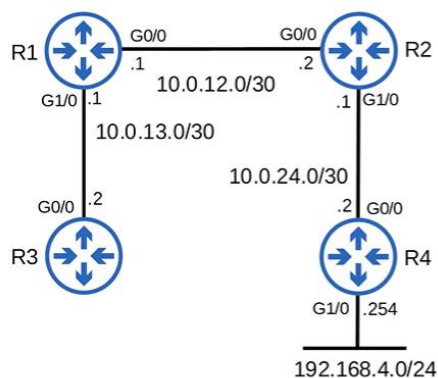


Dynamic Routing

- Topics covered:
 - Introduction To Dynamic Routing Protocols
 - Types of Dynamic Routing Protocols
 - Dynamic Routing Protocol Metrics

Dynamic Routing Protocols

Network Topology Used



Explanation:

Four routers: R1, R2, R3, R4.

LAN 192.168.4.0/24 is connected to R4.

Focus is mainly on R1.

R1 Routing Table (Before Routing Configuration):

Without static routes or dynamic routing protocols:

- Only connected routes
- Only local routes

```
10.0.0.0/8 is variably subnetted, 4 subnets, 2 masks
C    10.0.12.0/30 is directly connected, GigabitEthernet0/0
L    10.0.12.1/32 is directly connected, GigabitEthernet0/0
C    10.0.13.0/30 is directly connected, GigabitEthernet1/0
L    10.0.13.1/32 is directly connected, GigabitEthernet1/0
R1#
```

These are added automatically when IP addresses are configured.

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Network Routes vs Host Routes

Network Routes:

Examples:

- 10.0.12.0/30
- 10.0.13.0/30

Network route:

- Route to a network or subnet
- Mask length less than /32

Example:

- 192.168.4.0/24

Host Routes:

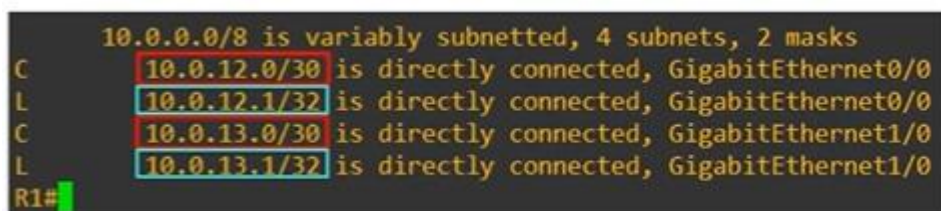
Examples:

- 10.0.12.1/32
- 10.0.13.1/32

Host route:

- Route to a specific host
- Mask length /32

These are automatically created for interface IP addresses.

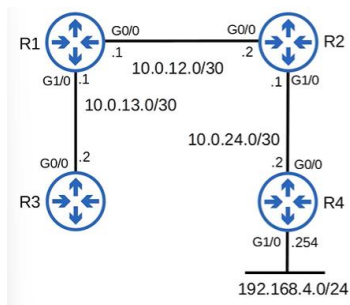


```
10.0.0.0/8 is variably subnetted, 4 subnets, 2 masks
C    10.0.12.0/30 is directly connected, GigabitEthernet0/0
L    10.0.12.1/32 is directly connected, GigabitEthernet0/0
C    10.0.13.0/30 is directly connected, GigabitEthernet1/0
L    10.0.13.1/32 is directly connected, GigabitEthernet1/0
R1#
```

Blue one is Host Routes & Red one is Network Routes.

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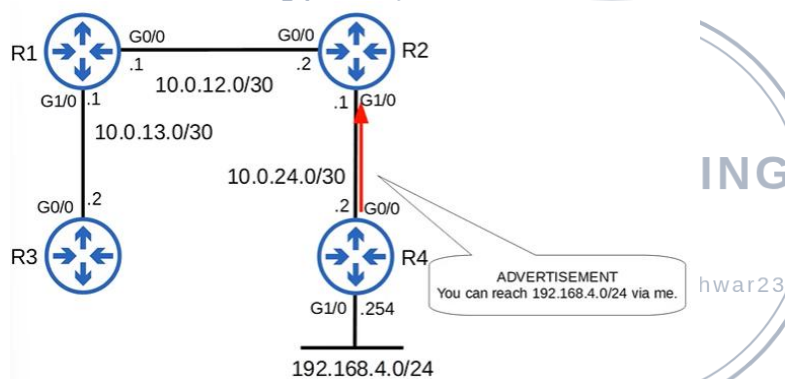
What Is Dynamic Routing



Instead of configuring static routes on each router, a dynamic routing protocol can be enabled.

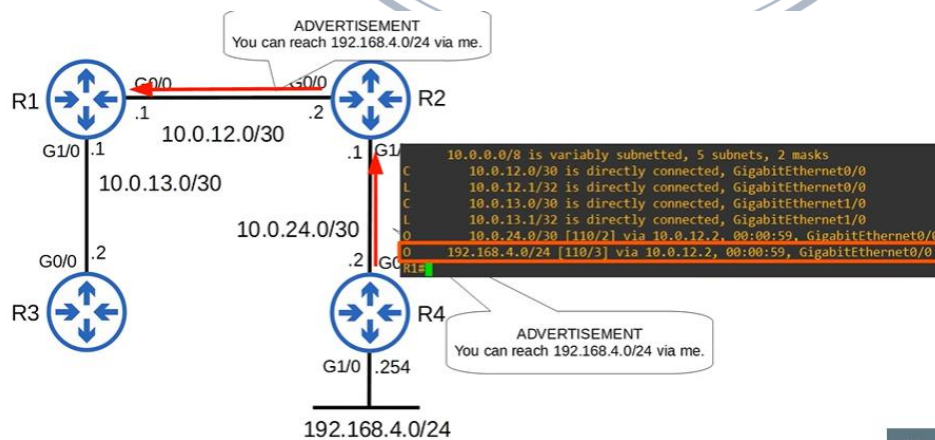
Route Advertisement Process

Step 1: R4 Advertises Its Network



Explanation: R4 tells R2 it can reach 192.168.4.0/24 via R4.

Step 2: R2 Advertises to R1



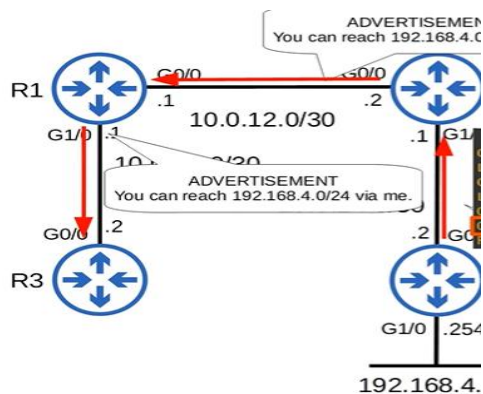
Explanation:

R2 adds the route and advertises it to R1.

R2 also advertises 10.0.24.0/30.

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Step 3: R1 Advertises to R3

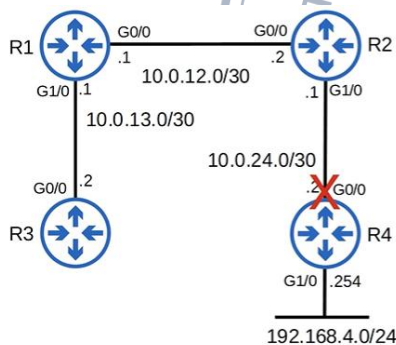


Explanation: R1 advertises the learned route to R3.

Automatic Route Removal

Failure Scenario:

R4's G0/0 interface goes down.



Result:

- Route to 192.168.4.0/24 is removed
- R1 stops sending traffic to a dead-end

```
10.0.0.0/8 is variably subnetted, 4 subnets, 2 masks
C    10.0.12.0/30 is directly connected, GigabitEthernet0/0
L    10.0.12.1/32 is directly connected, GigabitEthernet0/0
C    10.0.13.0/30 is directly connected, GigabitEthernet1/0
L    10.0.13.1/32 is directly connected, GigabitEthernet1/0
R1#
```

Explanation:

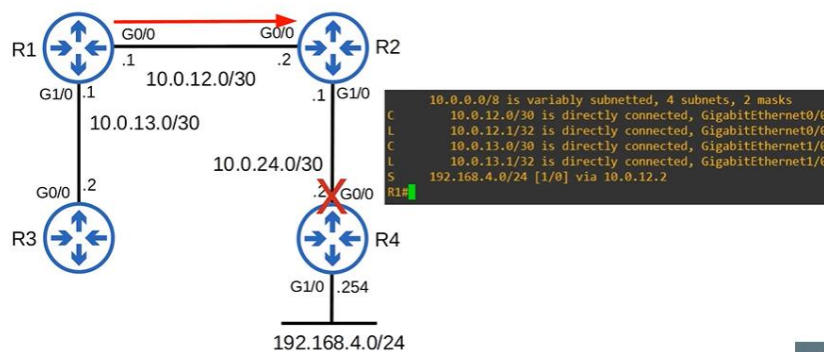
Dynamic routing automatically removes invalid routes.

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Static Routing Problem

With a static route on R1:

- R1 continues sending traffic
- R1 is unaware of the failure
- Packets are dropped

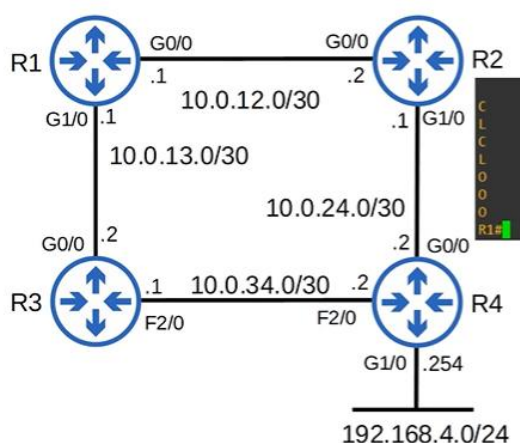


Benefit of Dynamic Routing

- Invalid routes are removed automatically
- Routers adapt to network changes

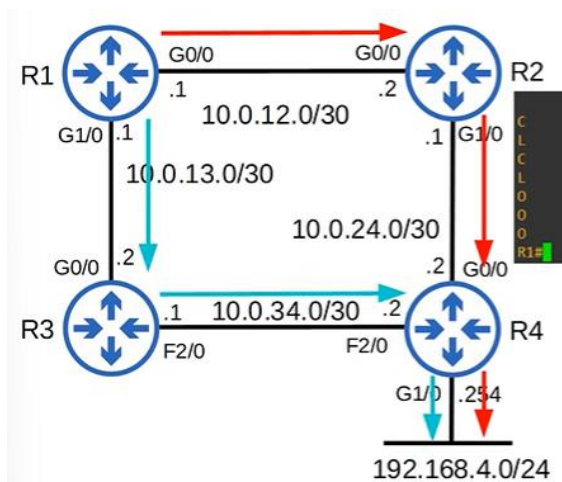
Backup Path Example

A new link is added between R3 and R4.



Explanation: R1 now has two paths to 192.168.4.0/24.

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Preferred Route Selection:

R1 prefers the route via R2.

```
10.0.0.0/8 is variably subnetted, 6 subnets, 2 masks
C    10.0.12.0/30 is directly connected, GigabitEthernet0/0
L    10.0.12.1/32 is directly connected, GigabitEthernet0/0
C    10.0.13.0/30 is directly connected, GigabitEthernet1/0
L    10.0.13.1/32 is directly connected, GigabitEthernet1/0
O    10.0.24.0/30 [110/2] via 10.0.12.2, 00:04:48, GigabitEthernet0/0
O    10.0.34.0/30 [110/11] via 10.0.13.2, 00:00:41, GigabitEthernet1/0
O    192.168.4.0/24 [110/3] via 10.0.12.2, 00:04:48, GigabitEthernet0/0
```

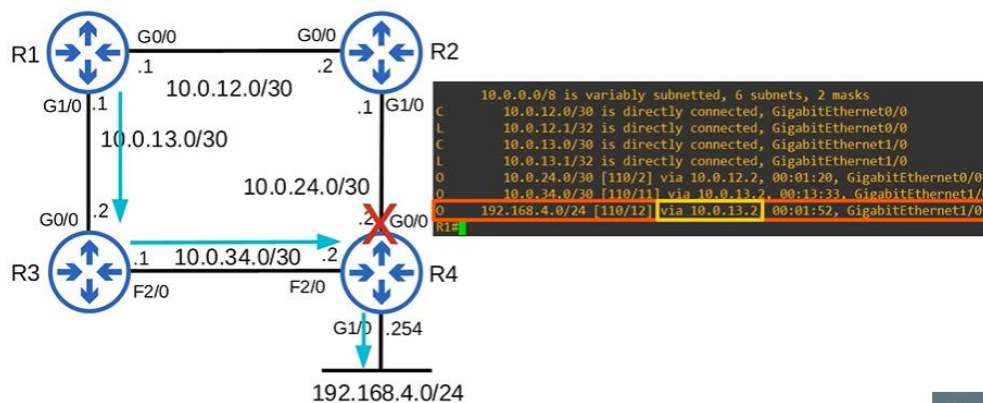
Reason:

- R2 path uses Gigabit Ethernet
- R3 path uses Fast Ethernet
- Lower cost path is preferred

Route Replacement After Failure:

When R4's G0/0 goes down:

- Route via R2 is removed
- Route via R3 is used



Computer Networks - CCNA

Metric Concept

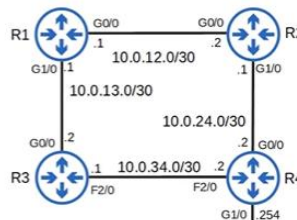
A metric measures how far a destination is.

- Used to select the best route
- Lower metric is better

Similar to root cost in spanning tree.

How Dynamic Routing Works

- Routers advertise routes
- Routers form adjacencies
- Best route is chosen using the metric
- Routers can use dynamic routing protocols to advertise information about the routes they know to other routers.
- They form 'adjacencies' / 'neighbor relationships' / 'neighborships' with adjacent routers to exchange this information.
- If multiple routes to a destination are learned, the router determines which route is superior and adds it to the routing table. It uses the 'metric' of the route to decide which is superior (lower metric = superior).



Types of Dynamic Routing Protocols

Dynamic routing protocols are divided into two types.

Interior Gateway Protocol (IGP)

- Used inside one autonomous system
- Example: a company network

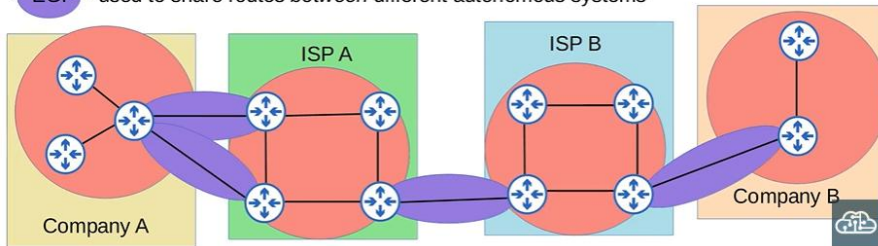
Exterior Gateway Protocol (EGP)

- Used between autonomous systems
- Used between companies and ISPs

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Autonomous System Example:

- IGP = used to share routes within a single *autonomous system* (AS), which is a single organization (ie. a company)
- EGP = used to share routes *between* different autonomous systems



Explanation:

- Each organization is an autonomous system
- IGP works inside an AS
- EGP works between ASs

CCNA Focus

- Main focus: OSPF (IGP)
- Other IGPs and the single EGP are covered to exam level

Key Takeaways

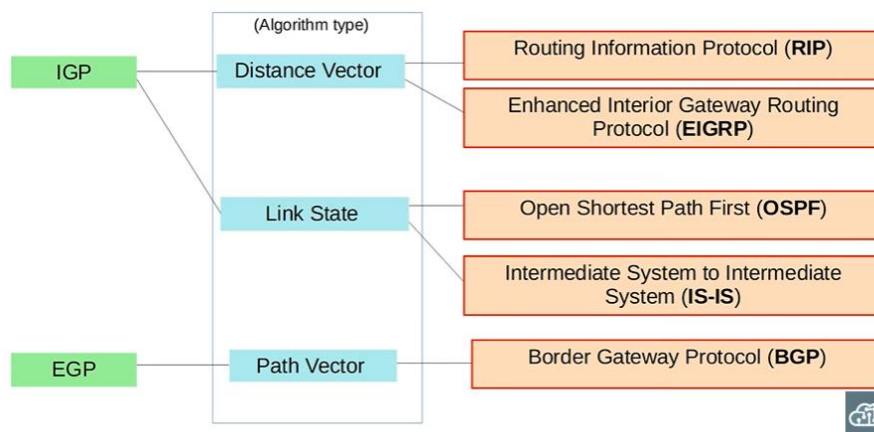
- Dynamic routing adapts automatically
- Routes are advertised between routers
- Best route uses the lowest metric
- Backup paths are selected automatically
- IGPs work inside an AS
- EGPs work between ASs

Types of Dynamic Routing Protocols

Routing Protocol Categories

Dynamic routing protocols are first divided into **IGP** and **EGP**, and then further divided by algorithm type.

Algorithm Type Overview:



Exterior Gateway Protocol (EGP)

- Only one algorithm type: Path Vector
- Only one EGP in use today: BGP

Purpose:

- Share route information between autonomous systems

CCNA Note:

- You do not need to know how Path Vector works
- Just know:
 - BGP is the only EGP
 - It is used between ASs

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Interior Gateway Protocols (IGP)

IGPs use two algorithm types:

- Distance Vector
- Link State

Distance Vector Routing Protocols

Distance vector protocols covered:

- RIP
- EIGRP

They were invented before link state protocols.

How Distance Vector Protocols Work:

Routers send the following to directly connect neighbours:

- Known destination networks
- Metric to reach those networks

This method is called routing by rumor.

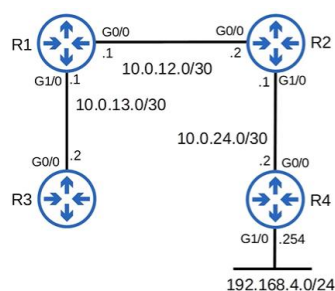
Why “Routing by Rumor”

- A router does **not** know the full network
- It only knows what neighbours tell it
- It trusts the information received

Distance + Vector Meaning:

- Distance = metric
- Vector = direction (next-hop router)

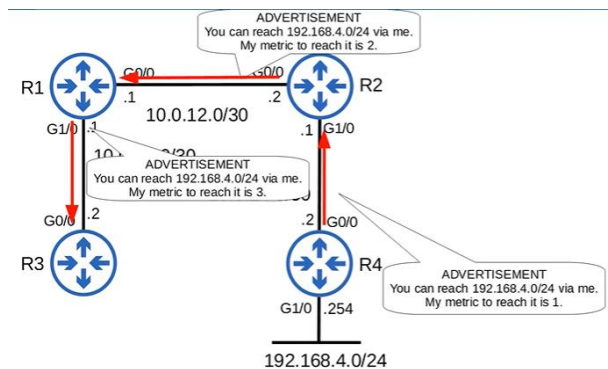
Example (Distance Vector Logic)



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Advertisements:

- R4 → R2: “192.168.4.0/24 via me, metric 1”
- R2 → R1: “192.168.4.0/24 via me, metric 2”
- R1 → R3: “192.168.4.0/24 via me, metric 3”



Each router only knows:

- Next hop
- Metric

Link State Routing Protocols

Link state protocols:

- OSPF
- IS-IS

How Link State Protocols Work:

- When using a **link state** routing protocol, every router creates a 'connectivity map' of the network.
- To allow this, each router advertises information about its interfaces (connected networks) to its neighbors. These advertisements are passed along to other routers, until all routers in the network develop the same map of the network.
- Each router independently uses this map to calculate the best routes to each destination.
- Link state protocols use more resources (CPU) on the router, because more information is shared.
- However, link state protocols tend to be faster in reacting to changes in the network than distance vector protocols.

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Link State vs Distance Vector

Distance Vector	Link State
Partial view	Full network map
Routing by rumor	Complete topology
Less CPU/memory	More CPU/memory
Slower convergence	Faster convergence



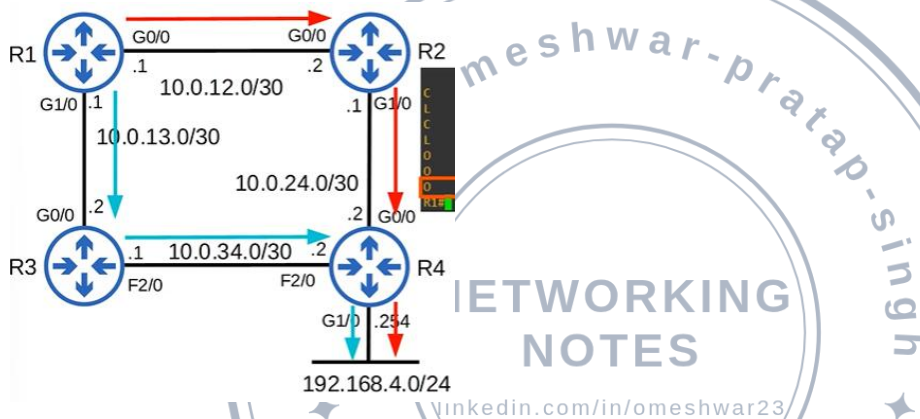
Dynamic Routing Protocol Metrics

Routing Metrics

A metric measures how good a route is.

- Used to choose the best route
- Lower metric = better route
- Like root cost in spanning tree

Route Selection Example:



R1 learns two routes to 192.168.4.0/24:

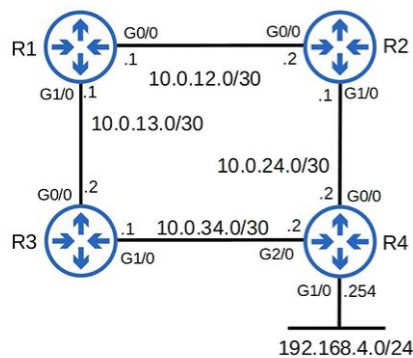
- Via R2 (Gigabit Ethernet) lower metric cost
- Via R3 (Fast Ethernet) higher metric cost

Only the route with the lower metric is installed, so in routing table only via R2 will show.

```
10.0.0.0/8 is variably subnetted, 6 subnets, 2 masks
C    10.0.12.0/30 is directly connected, GigabitEthernet0/0
L    10.0.12.1/32 is directly connected, GigabitEthernet0/0
C    10.0.13.0/30 is directly connected, GigabitEthernet1/0
L    10.0.13.1/32 is directly connected, GigabitEthernet1/0
O    10.0.24.0/30 [110/2] via 10.0.12.2, 00:04:48, GigabitEthernet0/0
O    10.0.34.0/30 [110/11] via 10.0.13.2, 00:00:41, GigabitEthernet1/0
O    192.168.4.0/24 [110/3] via 10.0.12.2, 00:04:48, GigabitEthernet0/0
R1#
```

Computer Networks - CCNA

Equal Cost Multi-Path (ECMP)



What If:

- Same routing protocol
- Same destination
- Same prefix length
- Same metric

Then:

- Both **routes are installed**
- Traffic is load-balanced

ECMP Example:

```
10.0.0.0/8 is variably subnetted, 6 subnets, 2 masks
C   10.0.12.0/30 is directly connected, GigabitEthernet0/0
L   10.0.12.1/32 is directly connected, GigabitEthernet0/0
C   10.0.13.0/30 is directly connected, GigabitEthernet1/0
L   10.0.13.1/32 is directly connected, GigabitEthernet1/0
O   10.0.24.0/30 [110/2] via 10.0.12.2, 00:00:09, GigabitEthernet0/0
O   10.0.34.0/30 [110/2] via 10.0.13.2, 00:00:09, GigabitEthernet1/0
O   192.168.4.0/24 [110/3] via 10.0.13.2, 00:00:09, GigabitEthernet1/0
    [110/3] via 10.0.12.2, 00:00:09, GigabitEthernet0/0
R1#
```

R1

└─ via R2 (metric 3)

└─ via R3 (metric 3)

Both routes appear in the routing table.

This is called ECMP (Equal Cost Multi-Path).

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ECMP in Routing Table:

```
R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2
        i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
        ia - IS-IS inter area, * - candidate default, U - per-user static route
        o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 6 subnets, 2 masks
C    10.0.12.0/30 is directly connected, GigabitEthernet0/0
L    10.0.12.1/32 is directly connected, GigabitEthernet0/0
C    10.0.13.0/30 is directly connected, GigabitEthernet1/0
L    10.0.13.1/32 is directly connected, GigabitEthernet1/0
O    10.0.24.0/30 [110/2] via 10.0.12.2, 00:00:09, GigabitEthernet0/0
O    10.0.34.0/30 [110/2] via 10.0.13.2, 00:00:09, GigabitEthernet1/0
O    192.168.4.0/24 [110/3] via 10.0.13.2, 00:00:09, GigabitEthernet1/0
    [110/3] via 10.0.12.2, 00:00:09, GigabitEthernet0/0
R1#
```

ECMP (Equal Cost Multi-Path)

(In Red Right side → Metric, In Blue left side → AD later in the course)

- Both routes show same destination
- Same metric value
- Load-balancing occurs

ECMP with Static Routes:

Static routes can also do ECMP.

- Two static routes to same destination
- Both installed
- Metric shown as 0
- Administrative Distance = 1

```
R1(config)#ip route 192.168.4.0 255.255.255.0 10.0.12.2
R1(config)#ip route 192.168.4.0 255.255.255.0 10.0.13.2
R1(config)#do show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2
        i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
        ia - IS-IS inter area, * - candidate default, U - per-user static route
        o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
        + - replicated route, % - next hop override
Gateway of last resort is not set

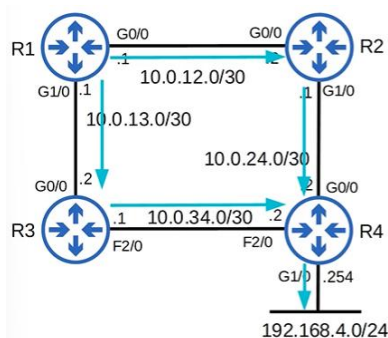
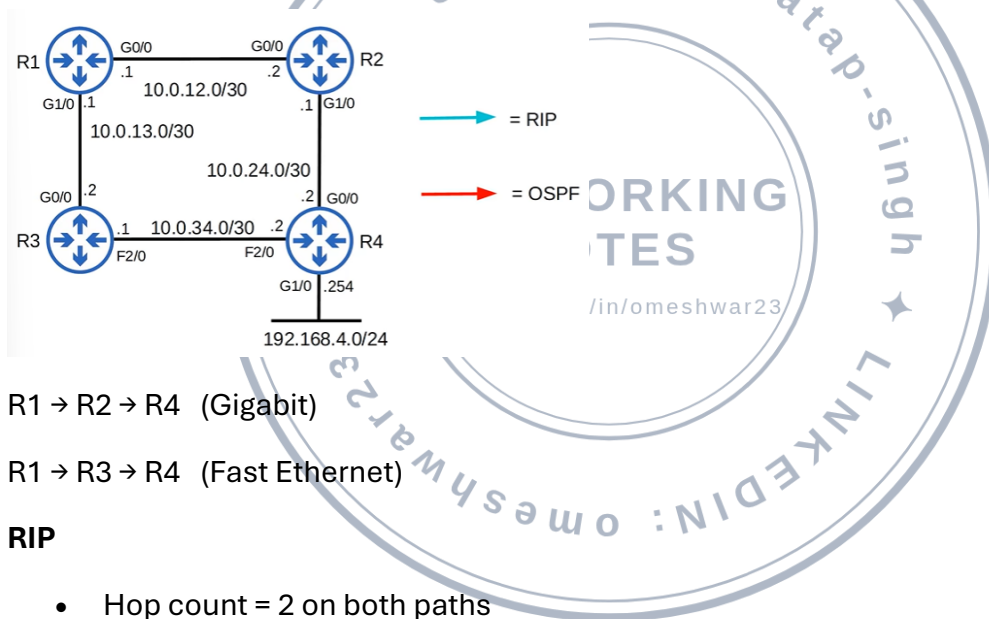
10.0.0.0/8 is variably subnetted, 4 subnets, 2 masks
C    10.0.12.0/30 is directly connected, GigabitEthernet0/0
L    10.0.12.1/32 is directly connected, GigabitEthernet0/0
C    10.0.13.0/30 is directly connected, GigabitEthernet1/0
L    10.0.13.1/32 is directly connected, GigabitEthernet1/0
S    192.168.4.0/24 [1/0] via 10.0.13.2
    [1/0] via 10.0.12.2
R1(config)#
```

Computer Networks - CCNA

Metrics Used by Common IGPs

IGP	Metric	Explanation
RIP	Hop count	Each router in the path counts as one 'hop'. The total metric is the total number of hops to the destination. Links of all speeds are equal.
EIGRP	Metric based on bandwidth & delay (by default)	Complex formula that can take into account many values. By default, the bandwidth of the slowest link in the route and the total delay of all links in the route are used.
OSPF	Cost	The cost of each link is calculated based on bandwidth. The total metric is the total cost of each link in the route.
IS-IS	Cost	The total metric is the total cost of each link in the route. The cost of each link is not automatically calculated by default. All links have a cost of 10 by default.

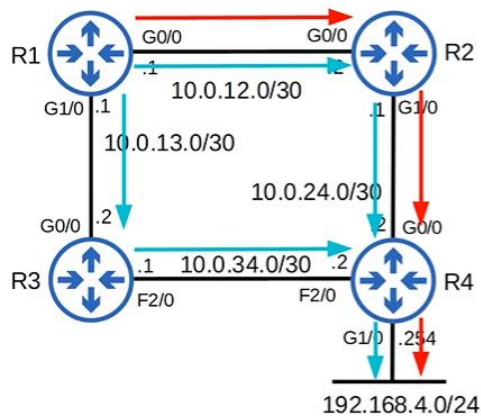
Metric Comparison Example (RIP vs OSPF)



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OSPF

- Slower link (via R3) = higher cost
- Only faster path installed (via R2)
- No load-balancing



Administrative Distance (AD)

- In most cases a company will only use a single IGP – usually OSPF or EIGRP.
 - However, in some rare cases they might use two. For example, if two companies connect their networks to share information, two different routing protocols might be in use.
 - Metric is used to compare routes learned via the same routing protocol.
 - Different routing protocols use totally different metrics, so they cannot be compared.
 - For example, an OSPF route to 192.168.4.0/24 might have a metric of 30, while an EIGRP route to the same destination might have a metric of 33280. Which route is better? Which route should the router put in the route table?
 - The **administrative distance** (AD) is used to determine which routing protocol is preferred.
- A lower AD is preferred, and indicates that the routing protocol is considered more 'trustworthy' (more likely to select good routes).

Metric compares routes learned by the same protocol.

Administrative Distance compares routes learned by different protocols.

- Lower AD = more trusted
- AD is checked before metric

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Common Administrative Distance Values (Cisco)

Route protocol/type	AD
Directly connected	0
Static	1
External BGP (eBGP)	20
EIGRP	90
IGRP	100
OSPF	110

IS-IS	115
RIP	120
EIGRP (external)	170
Internal BGP (iBGP)	200
Unusable route	255

If the administrative distance is 255, the router does not believe the source of that route and does not install the route in the routing table.

Important Notes About AD:

- AD is vendor-specific
- Routes with AD 255 are never installed
- Metric is only compared after AD

AD Quiz Example:

Routes to **10.1.1.0/24**:

- RIP via 192.168.1.1, metric 5
- RIP via 192.168.2.1, metric 3
- OSPF via 192.168.3.1, metric 10

Answer:

- OSPF route is selected
- Lower AD beats lower metric (Because OSPF is less AD compared to RIP)

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Reading the Routing Table

Format: [AD / Metric]

- Left value = Administrative Distance
- Right value = Metric

Example: [110/3] → OSPF route

```
10.0.0.0/8 is variably subnetted, 6 subnets, 2 masks
C    10.0.12.0/30 is directly connected, GigabitEthernet0/0
L    10.0.12.1/32 is directly connected, GigabitEthernet0/0
C    10.0.13.0/30 is directly connected, GigabitEthernet1/0
L    10.0.13.1/32 is directly connected, GigabitEthernet1/0
O    10.0.24.0/30 [110/2] via 10.0.12.2, 00:00:09, GigabitEthernet0/0
O    10.0.34.0/30 [110/2] via 10.0.13.2, 00:00:09, GigabitEthernet1/0
O    192.168.4.0/24 [110/3] via 10.0.13.2, 00:00:09, GigabitEthernet1/0
                        [110/3] via 10.0.12.2, 00:00:09, GigabitEthernet0/0
R1#
```

Example: [1/0] → Static route

```
10.0.0.0/8 is variably subnetted, 4 subnets, 2 masks
C    10.0.12.0/30 is directly connected, GigabitEthernet0/0
L    10.0.12.1/32 is directly connected, GigabitEthernet0/0
C    10.0.13.0/30 is directly connected, GigabitEthernet1/0
L    10.0.13.1/32 is directly connected, GigabitEthernet1/0
S    192.168.4.0/24 [1/0] via 10.0.13.2
                        [1/0] via 10.0.12.2
R1(config)#
```

Key Takeaways

- Algorithms define how routes are shared
- Distance vector = routing by rumor
- Link state = full network map
- Metric selects best route within same protocol
- ECMP allows load-balancing
- Administrative Distance selects between protocols
- Lower AD is always preferred

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Manual Changing Administrative Distance

You can change the AD of:

- A dynamic routing protocol
- A static route

This allows you to control which route is preferred.

Static Route with Default AD:

```
10.0.0.0/8 is variably subnetted, 4 subnets, 2 masks
C    10.0.12.0/30 is directly connected, GigabitEthernet0/0
L    10.0.12.1/32 is directly connected, GigabitEthernet0/0
C    10.0.13.0/30 is directly connected, GigabitEthernet1/0
L    10.0.13.1/32 is directly connected, GigabitEthernet1/0
S    192.168.4.0/24 [1/0] via 10.0.13.2
S    192.168.4.0/24 [1/0] via 10.0.12.2
R1(config)#
```

Explanation:

- This is a normal static route
- Default AD for static routes = 1
- Static routes are preferred over dynamic routes by default

Viewing Additional Options:

CLI Image: R1(config)# ip route 10.0.0.0 255.0.0.0 10.0.13.2 ?

```
R1(config)#ip route 10.0.0.0 255.0.0.0 10.0.13.2 ?
<1-255>  Distance metric for this route
multicast  multicast route
name       Specify name of the next hop
permanent  permanent route
tag        Set tag for this route
track      Install route depending on tracked item
<cr>
```

Explanation:

- The option <1-255> is shown
- This value is administrative distance
- It is not the routing protocol metric

Computer Networks - CCNA

Configuring a Static Route with Higher AD:

CLI Image: R1(config)# ip route 10.0.0.0 255.0.0.0 10.0.13.2 100

Explanation:

- The static route is configured with AD = 100
- Default static AD (1) is overridden

Verifying the Route:

```
R1(config)#ip route 10.0.0.0 255.0.0.0 10.0.13.2 100
R1(config)#do show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       + - replicated route, % - next hop override

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 6 subnets, 3 masks
S    10.0.0.0/8 [100/0] via 10.0.13.2
C    10.0.12.0/30 is directly connected, GigabitEthernet0/0
L    10.0.12.1/32 is directly connected, GigabitEthernet0/0
C    10.0.13.0/30 is directly connected, GigabitEthernet1/0
L    10.0.13.1/32 is directly connected, GigabitEthernet1/0
D    10.0.24.0/30 [90/3072] via 10.0.12.2, 00:06:35, GigabitEthernet0/0
R1(config)#
```

Explanation:

- The AD is now 100
- The metric is 0
- The AD value is shown inside the square brackets

Why Change the AD of a Static Route

By increasing the AD of a static route:

- It becomes less preferred
- Dynamic routing protocol routes can be preferred instead

But:

- The static route's AD must be higher than the dynamic protocol's AD

Computer Networks - CCNA

Floating Static Route

A static route with a higher AD is called a:

How It Works

- The route is not installed in the routing table
- It becomes active only if:
 - The dynamic route is removed
 - An interface failure occurs
 - A neighbour adjacency is lost

Summary (Quick Review)

- Dynamic routing protocols allow routers to learn routes automatically
- Static routes are useful, but not practical alone in large networks
- Routing protocols are divided into:
 - **IGPs** (within an organization)
 - **EGPs** (between organizations)
- The only EGP in use today is **BGP**
- Routing protocols use different **algorithms**:
 - Distance Vector: RIP, EIGRP
 - Link State: OSPF, IS-IS
- **Metrics** select the best route within the same protocol
- **Administrative Distance** selects between different protocols
- **Floating static routes** are static routes with higher AD used as backups